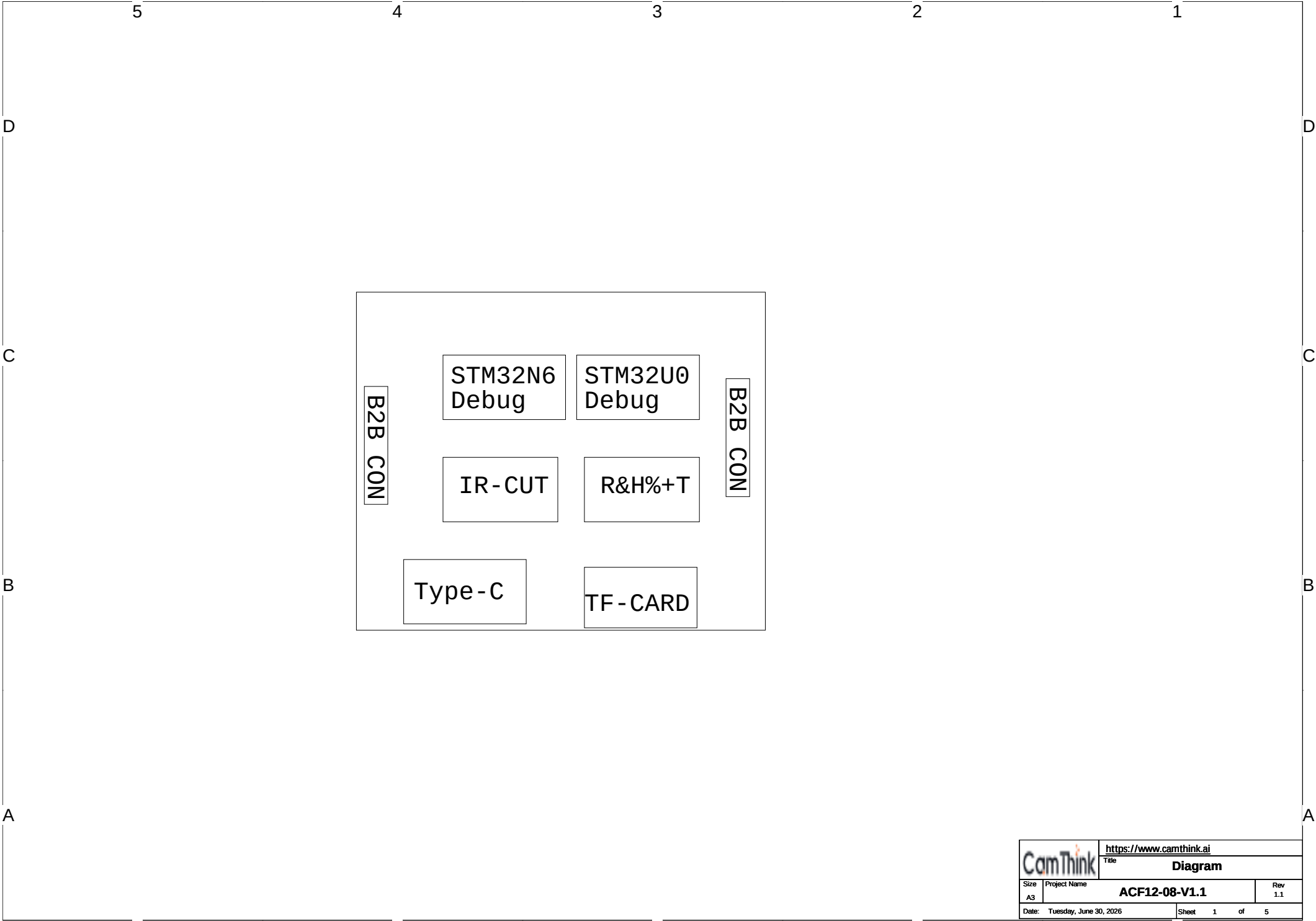




A

	https://www.camthink.ai		
	Title Revision		
Size A3	Project Name ACF12-08-V1.1		Rev 1.1
Date: Tuesday, June 30, 2026		Sheet 2 of 5	

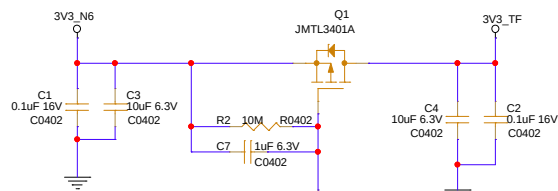
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4

3

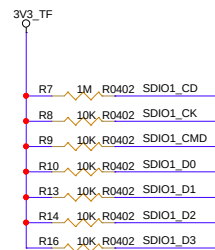
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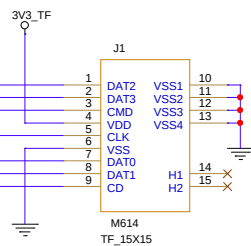
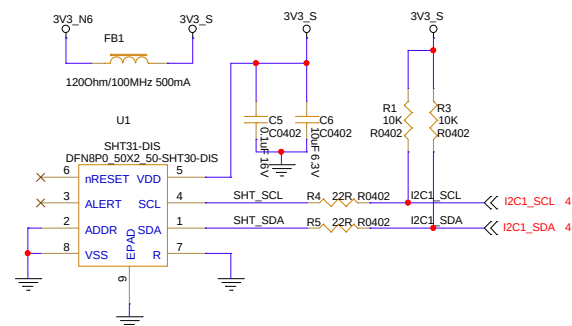
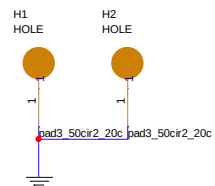
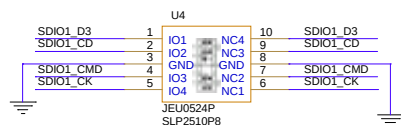
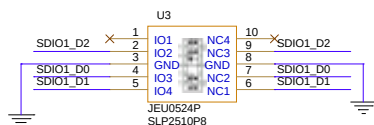


TF_PWR_ON=1:ACTIVE

4 TF_PWR_ON



4 SDIO1_D2
4 SDIO1_D3
4 SDIO1_CMD
4 SDIO1_CK
4 SDIO1_D0
4 SDIO1_D1
4 SDIO1_CD

M614
TF_15X15

I2C ADDR:
0x44 (Addr=0)
0x45 (Addr=1)

nRESET: (Input)
Active Low, if not used, left float or tied to VDD(R>2K)

Alert:
If not used, must be left float
Idle State: 0.2uA
Measuring: 1500uA Max

		https://www.camthink.ai	
Title		TF-Card & Sensor	
Size	Project Name	ACF12-08-V1.1	
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IR-CUT Driver

Top Section: Voltage Divider and Buffer

5V_VIN is connected to a voltage divider consisting of C23 (10uF 10V) and C24 (0.1uF 16V). The output of the divider is connected to the VIN pin of U6 (SG1301B50M). The VSS pin of U6 is connected to ground. The EN pin of U6 is connected to Boot1. The VOUT pin of U6 is connected to a voltage divider consisting of C25 (10uF 10V) and C26 (0.1uF 16V). The output of the divider is connected to 5V_IR.

Bottom Section: Motor Driver

SWF-12502WT is connected to a voltage divider consisting of R38 (10K) and R39 (1K). The output of the divider is connected to the VDD pin of U7 (AP1511B). The GND pin of U7 is connected to ground. The OUT1 pin of U7 is connected to the base of Q3 (MMBT3904). The OUT2 pin of U7 is connected to the emitter of Q3. The collector of Q3 is connected to 5V_IR. The emitter of Q3 is connected to ground. The IR_CUT_EN pin is connected to the base of Q3. The IR_CUT pin is connected to the collector of Q3. The IR_CUT pin is also connected to a voltage divider consisting of R40 (10K) and R41 (1K). The output of the divider is connected to IR_CUT_EN.

Truth Table:

FBC	OUT1	OUT2	OFF
1	1	0	1
0	0	1	0
Hold	Hold	Hold	Hold

200mA Max
Vin: 5.5V Max

